

BUPT SE ICP Quiz -- Unit 06 Arrays and Strings

If you are using LLM to help you in this quiz, you MUST write down 1) the name of the LLM and 2) the PROMPTS. Otherwise, you will get zero score if the instructor finds that you are using LLM.
You may use the following table as a template, putting the table down to the title of each question.

Name of LLM	Prompts
LongCat AI	1. I'm a beginner in learning CS, and I'd like to ask some questions 'bout the basic knowledge; 2. According to each question... // Questions solved with AI's help : Q5, Q7, Q14(AI improved)

- Define an float array coord[100][2] to store the x-y coordinates of 100 points , how to represent the coordinates of the 49th point? `float x = coord[48][0]; float y = coord[48][1];`;
- Define an float array coord[2][100] to store the x-y coordinates of 100 points , how to represent the coordinates of the 49th point? `float x = coord[0][48]; float y = coord[1][48];`;
- How to define a float array coords to store a 3d cube? `float coords[4][3];`;
// 从1个点延伸3个点, 记录每个点坐标
- Write the variable declaration and for loop necessary to create and initialize the following integer array:

```
int squares[11];
for(int i = 0; i <= 10; i++){
    squares[i] = i*i;
}
```

squares

0	1	4	9	16	25	36	49	64	81	100
0	1	2	3	4	5	6	7	8	9	10

- Define the following terms: bit, byte, word, address..

Name of LLM	Prompts
LongCat AI	1. I'm a beginner in learning CS, and I'd like to ask some questions 'bout the basic knowledge; 2. Define the following terms: bit, byte, word, address..

Bit: A bit (binary digit) is the smallest unit of data in computing and digital communications;

Byte: A byte is a unit of digital information that typically consists of 8 bits;

Word: A word is the natural unit of data used by a particular computer's architecture;

Address: An address is a unique identifier for a specific location in computer memory.

- Write a C expression that computes the number of bytes needed to represent an array containing **NElements** values of type **double**.

```
int NElements;
double Array[NElements];
```

```
int size = sizeof(Array);
```

7. What happens on most machines if you try to select an array element outside the allocated bounds of that array?

Name of LLM	Prompts
LongCat AI	1. I'm a beginner in learning CS, and I'd like to ask some questions 'bout the basic knowledge; 2.What happens on most machines if you try to select an array element outside the allocated bounds of that array?

In C, accessing an array element outside its allocated bounds results in **undefined behavior**.
 Typical Outcomes:Program Crash (Segmentation Fault);Silent Data Corruption;Security Vulnerabilities:

8. In you own words, state the Array Parameter Rule. From the programmer's point of view, what is the central difference between the operation of passing an array as a parameter and that of passing a simple variable?

The First parameter could be [] while the Array is stated, while others should be initialized (if there is); When passing a simple variable as a parameter, it's simply the pass of the value, and the value of the variable can not be changed; Yet while use Array, it could.

9. [T] True or False: When you declare an array parameter in a function, the array size is optional.
10. Assuming that the base address for the array is 1000 and that values of type int require four bytes of memory, draw a diagram that shows the address of each element in the array declared as follows: **int rectangular[2][3];**

[0][0]	[0][1]	[0][2]
1000	1004	1008
[1][0]	[1][1]	[1][2]
1012	1016	1020

11. Write a program that prompts the user to input a string and then prints the length of the string.

```
#include <stdio.h>
#include <string.h>

int main(void){
    char input[100];
    printf("Plz input a string\n");
    scanf("%99s",input);
    int stringLen = strlen(input);
    printf ("The lenth of the string is %d\n",stringLen);
    return 0;
}
```

12. 二维数组遍历（一重循环/嵌套循环）：Write a program that creates a 3x3 two-dimensional integer array and 1) uses a single loop to iterate through and print all elements, 2) uses nested loops to iterate through the array, and prints each element.

```
#include <stdio.h>
```

```
int main(void){
    int Array[3][3] = {
        {1,2,3},{2,3,4},{3,4,5}
    };
    for (int i = 1; i <= 9; i++){
        int col = i%3;
        int row = i/3;
        printf("%d",Array[col][row]);
    }
    for (int j = 0; j <= 2; j++){
        for (int k = 0; k <= 2; k++){
            printf("%d",Array[k][j]);
        }
    }
    return 0;
}
```

13. 查找元素：In a sorted one-dimensional array, write a program that finds and returns the position of a specified element.

```
#include <stdio.h>
```

```
int charPos(char element,char Array[]);
```

```
int main(){
    char spElement;
    char Array[10] = {'a','b','c','d','e','f','g','h','i','j'};
    scanf("%c",&spElement);
    int Pos = charPos(spElement,Array);
    if (Pos != 0) printf("The char you typed is in the %dth\n",Pos);
    else printf("404 NOT FOUND");
    return 0;
}
```

```
int charPos(char element,char Array[10]){
    for (int i = 0; i <= 9; i++){
        if (element == Array[i]){
            return i + 1;
        }
    }
    return 0;
}
```

14. 一维数组排序：Write a program that stores 10 integers using a one-dimensional array and sorts these integers in ascending or descending order.

```
#include <stdio.h>
```

```
void sort(int arr[], int n, int sortSort) {
    int i, j, temp;
    for (i = 0; i < n-1; i++) {
        for (j = 0; j < n-i-1; j++) {
            if (sortSort ? arr[j] > arr[j+1] : arr[j] < arr[j+1]) {
                temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
}

int main() {
    int arr[10], i;
    char sortSort;

    printf("Enter 10 integers:\n");
    for (i = 0; i < 10; i++) {
        scanf("%d", &arr[i]);
    }

    printf("\nWhich Sort of Sort wanna display (A/D): ");
    scanf(" %c", &sortSort);
    sort(arr, 10, (sortSort == 'A'));

    printf("\nSorted array:\n");
    for (i = 0; i < 10; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    return 0;
}
```

```
//AI improved Version
```

15. 字符数组的操作：Write a program that uses a character array to store a string, finds and replaces a specified character within the string.

```
#include <stdio.h>
```

```
int main(void){
    char Array[11] = "Hello,World";
    char repChar, repedChar;
    printf("Plz input 2 char, first must be in HelloWorld\n");
    scanf("%c %c", &repedChar, &repChar);
    for (int i = 0; i <= 10; i++){
        if (Array[i] == repedChar){
```

```
        Array[i] = repChar;
    }
    printf("%c",Array[i]);
}
printf("\n");
printf("All the %c has been replaced",repedChar);
}
```

16. Implement a function Capitalize (str) that returns a string in which the initial character is capitalized (if it is a letter) and all other letters are converted so that they appear in lowercase form. Characters other than letters are not affected. For example, Capitalize (“BOOLEAN”) and Capitalize (“boolean”) should each return the string “Boolean”.

```
#include<stdio.h>
#include <string.h>

void Capitalize(char str[],int len);

int main(){
    char input[100];
    printf("Plz type a string:");
    scanf("%s",input);
    int len = strlen(input);
    if (len > 0 && input[len-1] == '\n') {
        input[len-1] = '\0';
    }
    Capitalize(input,len);
    printf("%s\n",input);
    return 0;
}

void Capitalize(char str[],int len){
    str[0] -= 32; //用ASCII 尝试转大写
    for(int i = 1; i <= len -1; i++){
        if (str[i]<=93){
            str[i] += 32;
        }
    }
}
```



以下题目选做，不计入作业成绩

17. 二维数组的操作：Write a program that creates a 3x3 two-dimensional integer array, finds and prints the maximum and minimum values within the array.

```
#include<stdio.h>
```

```
int maxNum(int Array[3][3]);  
int minNum(int Array[3][3]);
```

```
int main(){  
    int Array[3][3] = {  
        {1,2,3},  
        {4,5,6},  
        {7,8,9}  
    };  
  
    int max = maxNum(Array);  
    int min = minNum(Array);  
  
    printf("The maximum num is %d \n",max);  
    printf("The minimum num is %d \n",min);  
  
    return 0;  
}
```

```
int maxNum(int Array[3][3]){  
    int max = Array[0][0];  
    for (int i = 0; i <= 2; i++){  
        for (int j = 0; j <= 2; j++){  
            if(Array[i][j] > max){  
                max = Array[i][j];  
            }  
        }  
    }  
    return max;  
}
```

```
int minNum(int Array[3][3]){  
    int min = Array[0][0];  
    for (int i = 0; i <= 2; i++){  
        for (int j = 0; j <= 2; j++){  
            if (Array[i][j] < min){  
                min = Array[i][j];  
            }  
        }  
    }  
    return min;  
}
```



18. 数组的实际应用：Write a program that simulates student score entry and output. Create a two-dimensional array to store the names of 10 students and their scores for 3 courses, and then print out the name of each student along with their average score.
19. Count the number of times the digits appear in the string, as well as the number of characters between 'e' and 'g'. **Input format:** The input is a string in a line. **Output format:** The output is two numbers separated by a space. The first is the number of all the digits, and the second is the number of characters between the two characters 'e' and 'g'. If there are multiple pairs of 'e'-'g', then the number corresponding to the pair with the largest gap between them will be output. **Sample input:** ee1g **Sample Output:** 1 2 **Data size and convention:** The input string consists of only uppercase and lowercase letters, numbers, and spaces. Spaces are also counted as one character. The total length of the string cannot exceed 3000 characters.